UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	PRE CALCULUS
COURSE CODE	MAT133
EXAMINATION	NOVEMBER 2026
TIME	2 HOURS

NSTRUCTIONS TO CANDIDATES

1. This question paper consists of four (4) questions.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) a one – page Appendix 1
   iii) an Answer Booklet - provided by the Faculty
   iv) a graph paper - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 5 printed pages

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QUESTION 1

a) Given that $f(x) = 2x^2 + 3$ and $g(x) = \sqrt{x+2}$.

i) State the domain of $f(x)$ and $g(x)$.
(2 marks)

ii) Determine $(f \circ g)(x)$.
(3 marks)

b) Given that $m(x) = \frac{2x+3}{2x-3}$, find the value of x if $m^{-1}(x) = x$.
(5 marks)

QUESTION 2

a) Solve the inequality $\frac{2}{3+2x} \leq -2$. Express the answer in interval notation.
(5 marks)

b) If $Z_1 = 1 - 2\mathrm{i}$ and $Z_2 = -3 + 5\mathrm{i}$, state $\frac{\overline{Z}_2 \mathrm{i}^{31}}{Z_1}$ in standard form $a + bi$.
(5 marks)

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QUESTION 3

a) Given two points $A(-4, p)$ and $B(2, 7)$ located on the equation $y = 2x + 3$. Find the value of p. Hence, find the equation of a straight line which is perpendicular to the line AB and passes through point A.
(5 marks)

b) Find the standard equation of circle where the diameter has endpoints $(-1, -8)$ and $(5, 0)$. Hence sketch the graph.
(5 marks)

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QUESTION 4

a) Given that $f(x) = x + 7$ and $g(x) = \frac{x+3}{x-4}$.

i) State the domain of $f(x)$ and $g(x)$
(2 marks)

ii) Determine $(g \circ f)(x)$. Hence, find $(g \circ f)(4)$.
(3 marks)

iii) Find the value of x if $(g \circ f)^{-1}(x) = 2$
(4 marks)

b) Use synthetic division to show that $(x - 1)$ is not a factor of $f(x) = 2x^3 - x^2 + 1$. Hence, state the quotient and remainder of $f(x)$.
(5 marks)

c) Find the value of x if $\frac{3^x - 9(3^{-x})}{2} = 4$.
(6 marks)

d) Based on the graph of $y = x^2$, sketch the graph of $y = -[(x+2)^2 - 1]$ by using transformation technique. Hence, state the x and y -intercept(s) (if any).
(5 marks)

END OF QUESTION PAPER

APPENDIX 1

Formula sheet (as supplied with the paper).

1. $\sin^2 A + \cos^2 A = 1$
2. $\sin(A + B) = \sin A\cos B + \cos A\sin B$
3. $\cos(A + B) = \cos A\cos B - \sin A\sin B$
4. $\tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A\tan B}$
5. $\cos 2A = \cos^2 A - \sin^2 A = 1 - 2\sin^2 A = 2\cos^2 A - 1$
6. $\sin 2A = 2\sin A\cos A$
7. $\tan^2 A + 1 = \sec^2 A$
8. $\tan 2A = \frac{2\tan A}{1 - \tan^2 A}$
9. Arc, $s = r\theta$
10. Area of sector $= \frac{1}{2}r^2\theta$
11. $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$
12. $a^2 = b^2 + c^2 - 2bc\cos A$
13. Area $= \frac{1}{2}ab\sin C$
14. Area $= \sqrt{s(s-a)(s-b)(s-c)}$

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